Implementation Science in Minnesota
An Informal Discussion Series

Today's Slides:
http://z.umn.edu/issoverview

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Welcome! What Are We Doing Here?

- Researchers at the U of MN, in conjunction with staff from the Minnesota Department of Education, are convening an informal discussion series on Implementation Science.
- What is Implementation Science?
  - Implementation research is the scientific study of methods to promote the systematic uptake of evidence-based practices into routine practice to improve the quality and effectiveness of services (Eccles & Mittman, 2006).
- Why Does Implementation Matter?
  - If an intervention or practice that has been demonstrated to be effective by research is not implemented properly with sufficient infrastructure to support fidelity to core effective components, the likelihood of success is low.

Objectives of this Series

- **Discuss current and future status of implementation science, including opportunities for research and training.**
- **Introduce common language and frameworks.**
- **Promote conversation and collaboration around a common topic.**
- **Identify actions and shared interests for research and practice development.**

Introductions

- Who is here today?
- Where do we work, what do we do, what areas of expertise are represented?
- Interested in learning more about implementation because…

Thank you for your time and interest!

Building Capacity for Effective Implementation Across Education

April 25, 2013
Minnesota Department of Education

MDE Implementation Team Members:
Eric Kloos, Cammy Lehr, Deborah Luedtke, Ellen Nacik, Debra Price-Ellingstad, Renee Ringold, Vicki Weinberg, Ruslana Westerlund, Elizabeth Ziemer

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National Center on State Implementation of Scaling-up of Evidence-based Practices (SISEP)

- June, 2008, Minnesota was one of four states selected to work actively with a new national technical assistance center
- Funded by United States Department of Education
- Based at the University of North Carolina
- Dean Fixsen, Karen Blase (U of North Carolina), George Sugai (U of Connecticut), Rob Horner (U of Oregon)
- Goal: *Building state capacity* for scaling up evidence-based practices

Capacity Development in Minnesota

- **Staff from SISEP provided ongoing technical assistance**
  - Monthly MN on-site visits from SISEP staff
  - Implementation Institutes
  - Website
  - Resource materials (briefs, binder, book)
  - Community of Practice (includes all 4 states)
- **SISEP intensive technical assistance continues**

What is "implementation" ?

- A specified set of activities designed to put into practice an activity or program of known dimensions (Fixsen et al. 2005)

What Are We Learning?

In its purest form, implementation is in service to well defined evidence-based programs and practices.

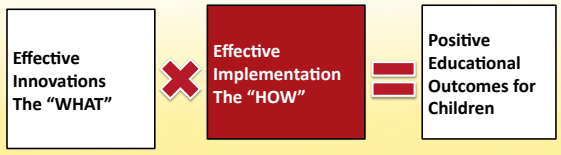
How are programs defined?

- Philosophy, values, & principles that underlie the program are described
- Criteria specifying who is most likely to benefit from the program must be described
- Essential functions that must be present are described (what "it" is/ is not)
- Operational definitions of the core components are clearly described (teachable, learnable, doable)
- Evidence that shows the program is effective when used as intended exists
- Performance assessments (fidelity) are highly correlated with outcomes

What Are We Learning?

- Sometimes we confuse educational programs or evidence based practices with funding sources, mandates or policies. e.g., Title I
- Implementation concepts can be applied to
  - Well Defined Programs – e.g., Check & Connect; Peer Assisted Learning Strategies (PALS)
  - Educational Frameworks – e.g., Positive Behavioral Interventions and Supports (PBIS)
  - Initiatives – e.g., MDE Educator Growth and Evaluation
- Everything interacts with everything else.
  - Programs, organizations, systems, policy, legislation, socio-economics are interdependent and interact dynamically
  - It gets messy
  - It is hard work and takes deliberate perseverance

Implementation Math: Multiplicative



Remember that any number times 0 is 0 !!!!

Students Cannot Benefit from Instructional Practices or Interventions That They Do Not Receive

What Are We Learning?

- An intervention (*the what*) is one thing.
- Implementation (*the how*) is something else altogether.
- We need both.
- Analogy: *the serum and the syringe*
  - Each has its own science and best practices
  - One without the other is ineffective

Minnesota's Common Principles of Effective Practice (CPEP) and Implementation

Working Together to Build Capacity for Effective Implementation of Effective Educational Practices

(see handout)

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Guiding Principles of the CPEP model

- Incorporates common principles of effective practice AND key components of implementation
- Not a mandate, offered as a way of work
- NOT another program – but a tool to help organize and integrate

Intended use as a…

- Framework to review educational initiatives and innovations
- Tool to plan professional development
- Lens through which to integrate programs and initiatives using a common language of the "what" and the "how"
- Reflection tool for assessing best practice
- Resource to gather basic information on implementation and effective practice

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What Are We Learning?

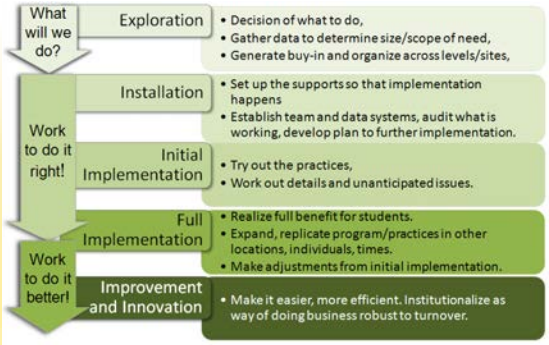
The research literature points to components that are key to effective implementation.

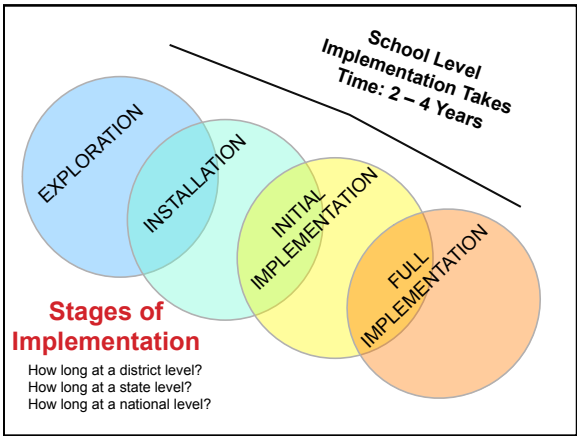
- **Stages of Implementation**
- **Implementation Drivers**
  - Staff Competence Drivers
  - Organization Drivers
  - Leadership at ALL Levels
  - Performance Assessment

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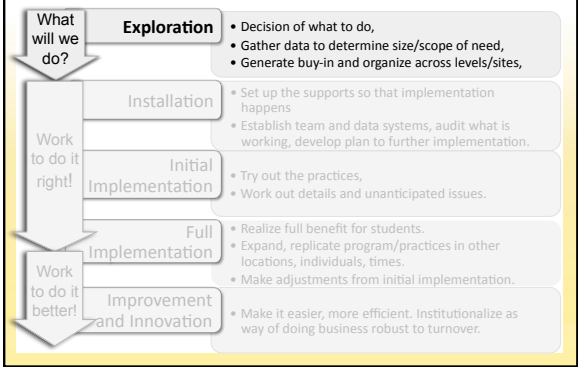
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Component 1: **Implementation Stages**





Component 1: **Stages of Implementation**

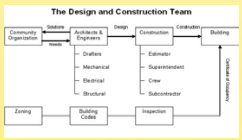


EXPLORATION: Creating the Blueprint

- What is the function the space must serve
- What is the scope of the project:
  - Remodeling
  - Restoration
  - Additions
  - New Construction



- What is timeline?
- What are expected resources we have/need?
- What is the cost?
- What is our staffing/talent pool?



EXPLORATION: Creating the Blueprint

Who needs to be in on the planning?



What is the scope of the project?

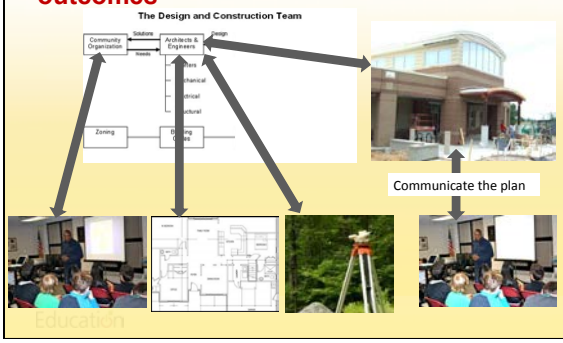
- What results are we getting with our current efforts? Needs?

What do we still need to achieve the goal?



- What does our data tell us? Are we missing data?
- What do we add/problem solve/drop/make room for?
- What can we manage with time, resources, and available money?

EXPLORATION: Anticipating what is needed to manage project towards desired outcomes



Reflections…

What will we do?	**Exploration**	• Decision of what to do, • Gather data to determine size/scope of need, • Generate buy-in and organize across levels/sites,

- **Discuss the importance of Exploration.**
- **How do you reconcile the pressure to skip exploration with its importance?**
- **How do know when you are done with exploration?**
- **What are the indicators that you need to revisit exploration?**

Component 1: **Stages of Implementation**

What will we do?	Exploration	• Decision of what to do, • Gather data to determine size/scope of need, • Generate buy-in and organize across levels/sites,
Work to do it right!	**Installation**	• Set up the supports so that implementation happens • Establish team and data systems, audit what is working, develop plan to further implementation.
	Initial Implementation	• Try out the practices, • Work out details and unanticipated issues.
	Full Implementation	• Realize full benefit for students. • Expand, replicate program/practices in other locations, individuals, times. • Make adjustments from initial implementation.
Work to do it better!	Improvement and Innovation	• Make it easier, more efficient. Institutionalize as way of doing business robust to turnover.

Installation: Laying the Foundation

Installation IS:

- Our **data is telling us** that we have **a cohort** of first grade students who **need**…
- Therefore next year we will:
  - Allocate X resources for small groups through Fall
  - 3 teachers will get X training
  - 2 teachers get X coaching
  - We will **set a date to revisit** student progress in winter

Installation: Laying the Foundation…

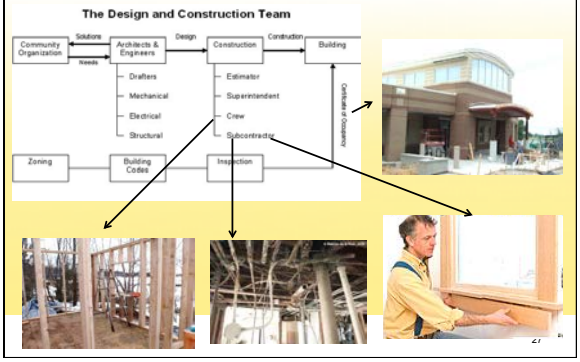
Installation IS NOT:

- Writing a plan
- Submitting data
- We have a manual, a person, a room
- Reporting student progress to parents
- Faculty were trained= 20 minute meeting communicating the plan
- Saying "We are doing it" in the absence of data or evidence of "doing it well"

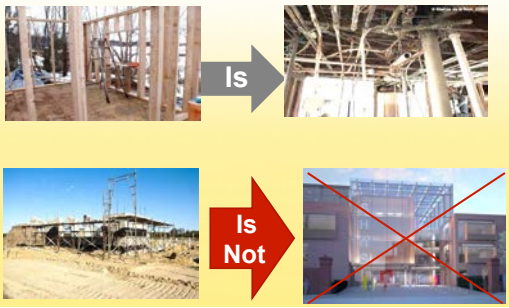
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Initial to Full Implementation



Initial and Full Implementation:



Implementation: Actively Manage Drivers

You are likely to have to :

- Continue to review data, collect effort and fidelity data
- Examine outcomes – are we getting intended impact – make necessary changes
- Continue to develop an organization that is a good host for managing the work
- Manage staff competency (coaching and training)
- Problem solve and manage adaptive and technical challenges.



Component 1: Stages of Implementation

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Innovation: Is Driven by Data After We Have Reached Full Implementation

- Innovation comes from seeing how benefit from existing practices/implementation
- Innovation is about creating efficiencies







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Innovation: Is and Is Not…



Is





Is Not



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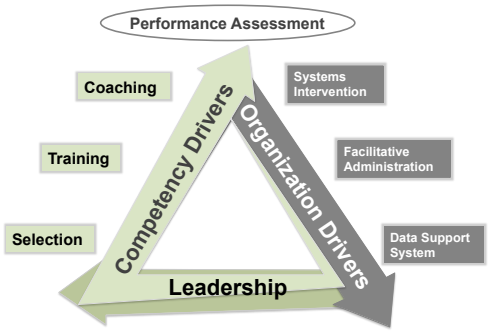
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Reflections…

- How do reconcile the pressure to skip stages?
- Discuss the importance of fully implementing a practice before innovating.
- How long does this process take?
- What are some potential benefits of applying stage based implementation framework to your work?

Work to do it better! | Improvement and Innovation | • Make it easier, more efficient. Institutionalize as way of doing business robust to turnover.

Component 2: **Implementation Drivers**



IMPLEMENTATION DRIVERS

- Common features of successful supports to help make full and effective use of a wide variety of well defined educational practices, programs and initiatives.
- Implementation drivers are the core components of change at teacher, school, and district levels.
- The drivers represent a way to think about plan for, and evaluate implementation activities within the organization.
- The drivers are integrated and compensatory.

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Implementation:
Benefits of Driver-Based Planning

- Infrastructure needed becomes visible to all
- Strengths and progress get celebrated
- Next right steps are planned and results measured
- Resources can be aligned and re-purposed to improve implementation

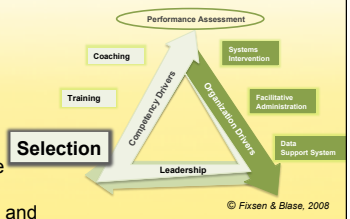
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Implementation Drivers:

Ways to select:
- Hiring
- Re-assigning
- Recruiting

Critical Features:
- Apply critical skills
- Includes non-negotiable competencies
- Assess ability to accept and incorporates feedback



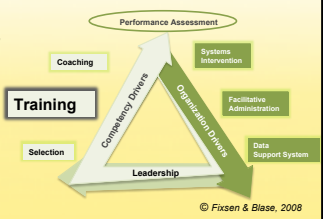
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Implementation Drivers:

Program
- Key components
- Background knowledge
- Rationale of key practices
- Applied practice of skills

Critical Features
- Critical skills
- Solve real life problems
- Behavioral rehearsals
- Provide demonstrations
- Practice and feedback



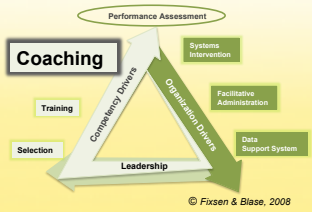
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Implementation Drivers:

Selection Criteria
- Competency to support adult learners
- Ability to establish trust and rapport

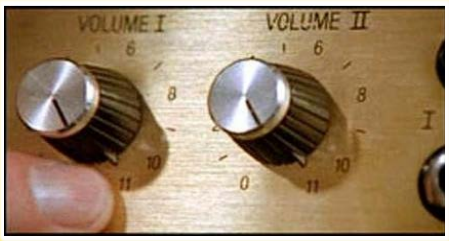
Critical Features
- In service to training
- Include service delivery plan
- Include observational component



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Integrated and Compensatory



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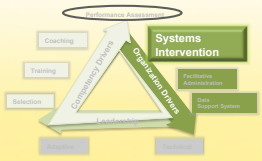
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Reflection…

- Think of a time that you used one or more selection, training and coaching to strengthen implementation. Did you have to compensate for one or two of the competency drivers?
- How might you be more purposeful in strengthening one of the following to improve implementation?
  - **Selection**
  - **Training**
  - **Coaching**
- How could you use staff competency drivers more purposefully than how you are currently using them?

Organization Drivers: Systems Intervention

Definition:
Strategies to work with external systems to ensure the availability of the financial, organizational, and human resources required to support the work of the practitioners.



Systems Trump Programs!

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Organization Drivers: Systems Intervention

Used:

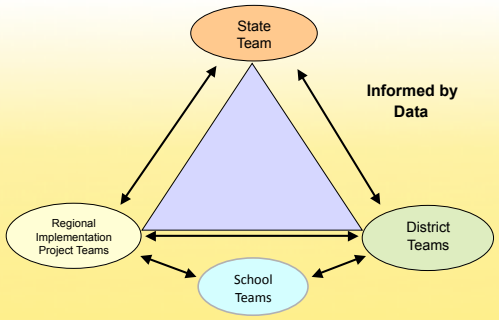
- When issue is outside your sphere of influence
- To communicate up one level

Purpose:

- Identify barriers and facilitators
- Create a "hospitable" environment and buy-in across levels
- Facilitate learning
- Enact supportive policy

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Linked Implementation Teams



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Organization Drivers: Facilitative Administration

To Support Implementation:

- Facilitates moving through the stages
- Ensures the effective use of the competency drivers
- Takes the lead on Systems Interventions
- Utilize improvement cycles

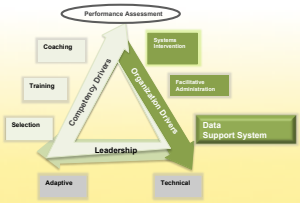


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Organization Drivers: Data Support Systems

Uses

- Provide information to assess effectiveness of new practices
- Celebrate successes
- Inform staff and families
- Engage in continuous quality improvement
- Be accountable for quality infrastructure (are the drivers working) and for outcomes



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Organization Drivers: Data Support Systems

Reminder

- All data should serve a purpose
- Collect data with fidelity
- Be prompt about looking at data and acting on it
- Data should be at an actionable level
- Use multiple sources of data to confirm what you see

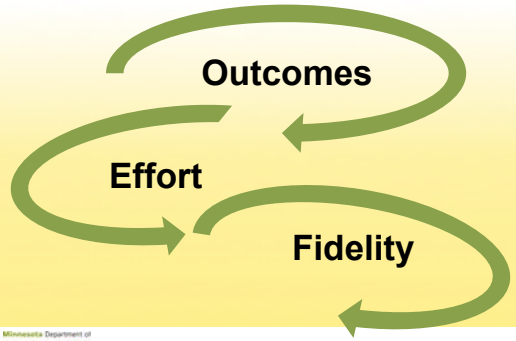
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Reflections…

- What are some of the external systems that influence your implementation efforts? What are strategies to increase the likelihood of successful implementation of your innovation?
- How can facilitative administration help you navigate through the stages of implementation?
- Which type(s) of data are most problematic to collect, analyze, and use? Why?

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Evidence of Performance Data



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Using Performance Data

Student Outcomes

- Are we reaching our desired outcomes? Is our data demonstrating progress towards the intended outcome?

Not Yet

Effort

- What is the evidence telling us about our efforts? Which of the defined actions occurred?
- Is there a lot of effort or only a little? Is it widespread or only in some areas?

A lot

Fidelity

- How consistently are our efforts being carried out? Are they as described in our plan?
- Do our efforts need to change to become more effective?

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Data Informs Active Management of Other Drivers



Reflections…

Reflect upon a program or practice you are currently implementing.

Outcomes: How are student outcomes being measured? Are you reaching desired outcomes?

Efforts: How can the implementation efforts be measured? What information can be used to tell you about the implementation efforts?

Fidelity: How can the fidelity of program implementation be measured? How well are the components of the "WHAT" being implemented?

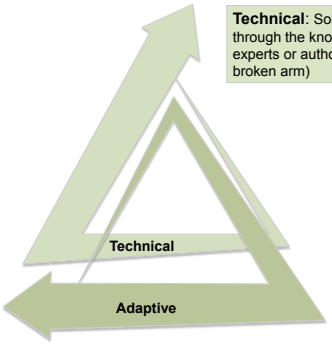
Gaps and Actions: Are all of these pieces of information typically collected? Why or why not?

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Leadership Driver

Adaptive: Solve outside your current toolkit or repertoire; it consists of a gap between aspirations & operational capacity that cannot be closed by the expertise and procedures currently in place.
-Ronald Heifetz

Technical: Solve through the knowledge of experts or authority (e.g., broken arm)



Leadership v. Authority

"….for too long we have been training leaders who only know how to keep the routine going. Who can:

…answer questions, but don't know how to ask them.

…fulfill goals, but don't know how to set them.

…think about how to get things done, but not whether they're worth doing."

-William Deresiewicz

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What leadership support might look like…

- Make public statements of support for the educational practice or program
- Establish, written or otherwise, implementation of the educational practice or program is a top priority
- Motivate staff to take up the charge
- Allocate resources
- Participate in process planning or implementation activities

Lohrmann, S., Forman, S., Martin, S., Palmieri, M., (2008). Understanding School Personnel's Resistance to Adopting Schoolwide Positive Behavior Support at a Universal Level of Intervention. *Journal of Positive Behavior Interventions, 10(4), 256-269.*

Implementation Science

www.scalingup.org
http://sisep.fpg.unc.edu
State Implementation and Scaling-up of Evidence-based Practice

Implementation Research:
A Synthesis of the Literature



Fixsen, D. L., Naoom, S. F., Blase, K. A., Friedman, R. M. & Wallace, F. (2005). *Implementation Research: A Synthesis of the Literature.* Tampa, FL: University of South Florida, Louis de la Parte Florida Mental Health Institute, The National Implementation Research Network (FMHI Publication #231).
Download all or part of the monograph at:
http://www.fpg.unc.edu/~nirn/resources/detail.cfm?resourceID=31

Join Us for Next Sessions

- **May 2: Emerging Issues in Research and Development – The Conceptual and Empirical Base for Implementation Science**
  - **Scott McConnell, Tracy Bradfield, and Laura Potter, UM**
- **May 16: Implementation Science Today and in the Future: Local and National Examples**
  - **Presenters from MDE, UM, others (volunteer!)**
- **May 30: Future Research, Development, and Use of Implementation Science**
  - **Cammy Lehr (MDE) and Scott McConnell (UM)**

Thanks for coming!